PROCESS-2 corpus

400 participants

HC	200
MCI	150
Dementia	50

3 tasks, one session

- **SFT** semantic fluency (animals, 60 s)
- **PFT** phonemic fluency (letter P, 60 s)
- **CTD** picture description (Cookie Theft)

audio (16 kHz)

3 s

manual transcripts

official split

train 320 | 80 test

colour key

- annotation shortcut
- annotation-free model
- HC
- MCI
- dementia

A The problem: the same speech is transcribed in two ways, and the way depends on diagnosis

Illustrative example (not corpus data): five animals named in the semantic fluency task

What the participant says



"cat … dog, lion, horse, cow"

3-s silence after "cat"

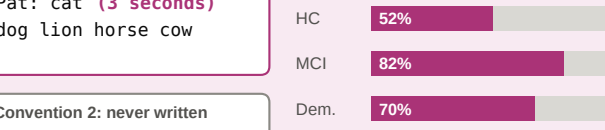
Convention 1: pauses written

```
Pat: cat (3 seconds)
dog lion horse cow
```

Convention 2: never written

```
cat dog lion horse cow
```

Sessions with convention 1



Missing annotations are not missing pauses: HC pause as much in the audio

A text model can learn:

"seconds" → impaired

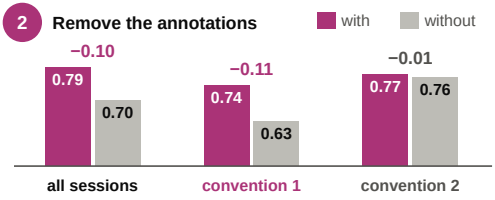
B The audit: is the gain due to the annotations or to the speech?

1 What does the model read?

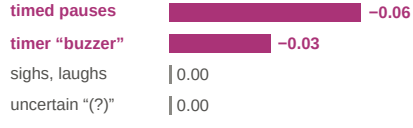


"seconds" is the top word in 3/3 tasks and 25/25 folds

2 Remove the annotations



3 Which annotations matter?



AUC lost when one category is removed: two carry it

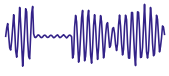
4 Other checks

- demographics, channel **weak (AUC ≤ 0.56)**
- silence alone to audio **≈ amount of pausing**
- zero-shot LLMs **no prior use**
- cleaned transcripts **other habits remain**

C The control: a model that never reads a human annotation

Same example; every step is automatic and identical for all participants (values illustrative)

Input



audio only

no transcript, no annotation

1 Voice activity detection



→ silences measured in the audio
→ **median pause, long silences**

2 Speech recognition

```
cat dog lion horse cow
```

→ 5 valid animals in 45 s
→ **fluency = 0.11 words/s**

3 Healthy norms



compare with healthy people of same age, sex

fluency z = −1.5
pauses z = +1.2

(6 measures in total)

4 Risk score



6 visible weights

fewer words and longer pauses
→ **higher risk**

Official test set, evaluated once: AUC 0.81 [0.70, 0.91]

same under both transcription conventions
all choices made by cross-validation on the training set